

Dec-24-0078

MAFC-311 (Probability Theory and Statistics)

[EEE, ECE, CSE, AI&ML, CSE (AI&DS)]

B.Tech, 3rd

Time : 3 Hours

Max. Marks : 60

Note : The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all, one question each from sections A, B, C and D. Section E is compulsory. Use of non-programmable calculator is allowed.

SECTION - A

1. (a) From a lot of 12 items containing 3 defective items, a sample of 4 items is drawn at random without replacement. Let a random variable X denote the number of defective items in the sample. Find the probability distribution of X . **(6)**

(b) A continuous random variable X has pdf $f(x) = 6x(1-x)$, $0 \leq x \leq 1$. Determine a number b such that $P(X \leq b) = P(X > b)$. **(6)**

OR

2. (a) Two players A and B play tennis games. Their chances of winning a game are in the ratio 3:2 respectively. Find the chance of A winning at least two games out of four games played. **(6)**

(b) [A second continuous-random-variable question was on this page but was not legible in the photo — please refer to your original.] **(6)**

SECTION - B

3. (a) Fit a Poisson distribution to the following data and test its goodness of fit at 5% level of significance: $x = 0, 1, 2, 3, 4$ with frequencies 419, 352, 154, 56, 19 (value of χ^2 for 3 degrees of freedom at 5% level of significance is 7.82). **(6)**

(b) Two large samples of sizes 9 and 8 give sums of squares of deviations from their respective means equal to 160 inches² and 91 inches² respectively. Can these be regarded as drawn from the same normal population? (given that F critical ≈ 3.73 at 3 and 7 degrees of freedom at 5% level of significance). **(6)**

OR

4. (a) Given a table of x and $f(x)$ values, find the Lagrange interpolating polynomial that fits the data and use it to estimate $f(x)$ at the required point. [Exact table values were not fully legible in the photo — please refer to your original.] **(6)**

(b) Evaluate $f(19)$ using Newton's divided difference formula from the given table of values. **(6)**

SECTION - C

5. (a) Using three iterations of the Newton-Raphson method, find an approximate root of the given equation, using a suitable initial approximation. **(6)**

(b) Given a table of x and $f(x)$ values, evaluate $f(x)$ at the required point using Newton's divided difference formula. **(6)**

OR

6. (a) Find the Lagrange interpolating polynomial that fits the given data values and determine the approximate value of $f(x)$ at the required point. **(6)**

(b) A root of the given equation lies in a specified interval. Determine this root correct to three decimal places using the regula-falsi method. **(6)**

SECTION - D

7. (a) Using Simpson's rule, evaluate the given definite integral with the specified number of nodal points. **(6)**

(b) Given an initial value problem $dy/dx = f(x,y)$, $y(x_0) = y_0$, find an approximate value of y at the required point using the modified Euler's method. **(6)**

OR

8. (a) Solve the given initial value problem in the specified interval using the Runge-Kutta fourth order method with the given step size. **(6)**

(b) Obtain the approximate value of y at the required point for the given initial value problem using the Taylor series method / second order method with the specified step size. **(12)**

SECTION - E

(Compulsory)

9. [Section E of this paper (short compulsory questions, typically worth 2 marks each) was not clearly captured in the provided photographs — please refer to your original booklet.] **(6×2=12)**

Transcribed from a photograph of the original answer/question booklet. A few numeric values, table entries and one section could not be fully verified due to image quality (angle/blur/overlap with adjacent papers) — please cross-check against your original copy. Section E and several numeric tables in this paper were especially hard to read in the photo and have been left as placeholders — please fill these in from your original.